Computation Practical 9: Phylogenetics Visualisation

Exercise 1: Investigating an outbreak of CPE cases

Introduction

In this practical we will investigate the **CPE cases** we have collectively collected data for during the course. We will specifically focus on the ST78 *Klebsiella pneumoniae* isolates of the CPE cases. We will integrate the epidemiological and antibiotic susceptibility data collected using **EpiCollect** in previous practicals, along the with the phylogenetic tree we generated. We will also contextualise our local hospital cases in relation to other isolates collected from other regions in the world.

We will make use of **Microreact** (https://Microreact.org/), a web application that provides an interactive visualization of datasets via phylogenetic trees, maps, timelines, and tables. But first we will download the epidemiological data collected using Epicollect to draw conclusions on the origin and spread of the CPE outbreak cases.

Download collected data from Epicollect

Visit the following URL to visualise the data collected by all participants:

https://five.epicollect.net/project/south-africa-cpe-outbreak

Login using the provided credentials to access the project main page (Figure 1). Click on 'VIEW DATA' to access all entries in this project.

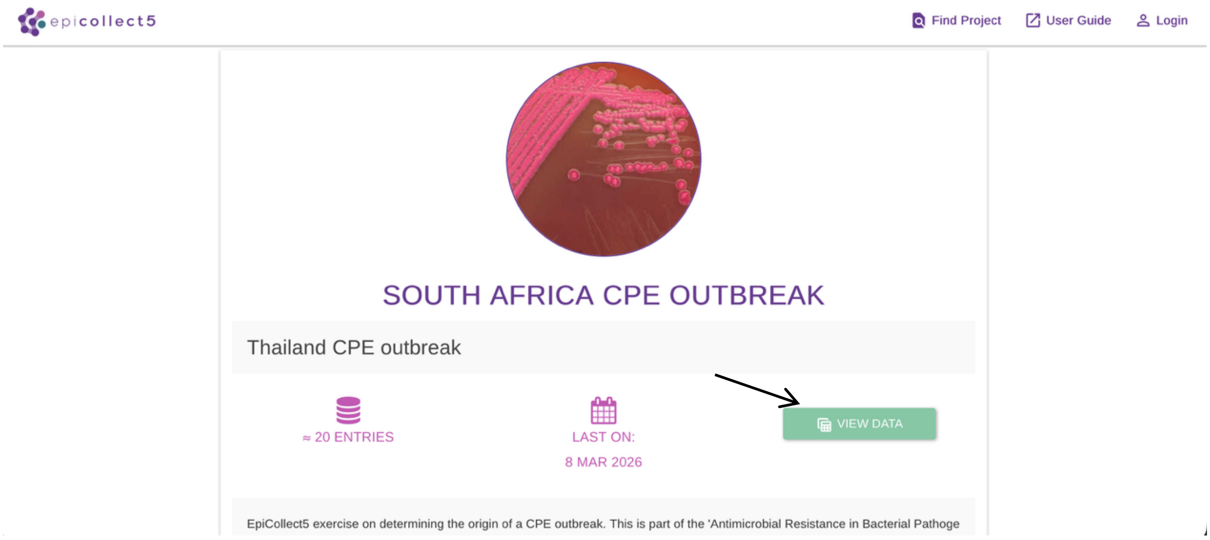

Figure 1 South Africa CPE outbreak EpiCollect project page

You will see then the data collected by all course attendees as a table (Figure 2).

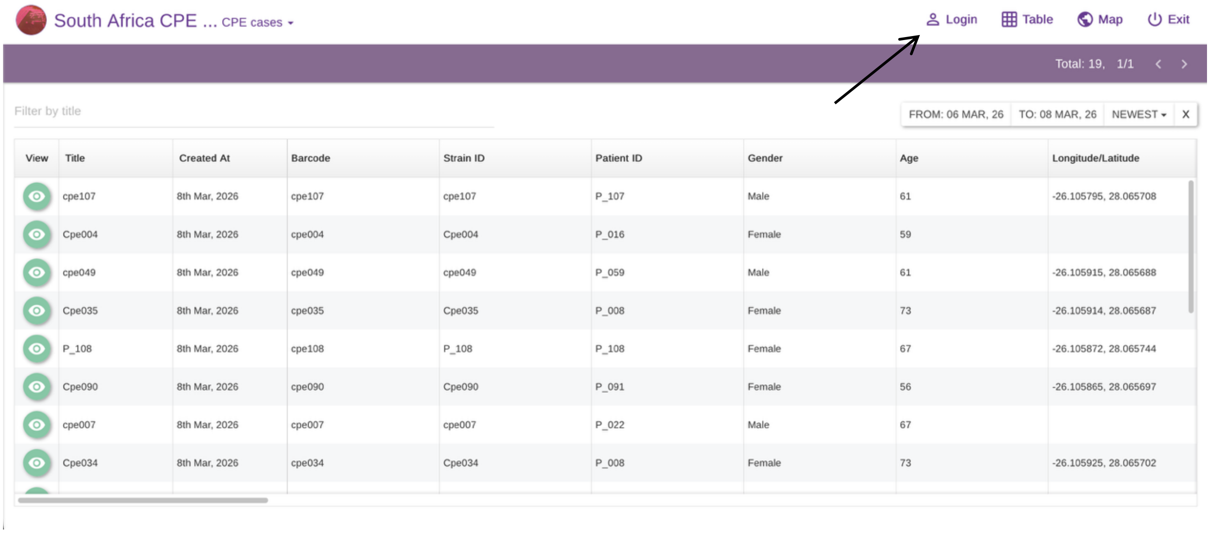

Figure 2 South Africa CPE outbreak EpiCollect project data

To be able to download this data locally we need to Login first (arrow in Figure 2).

Click on 'Download' (Figure 3) to download all entries as a CSV file.

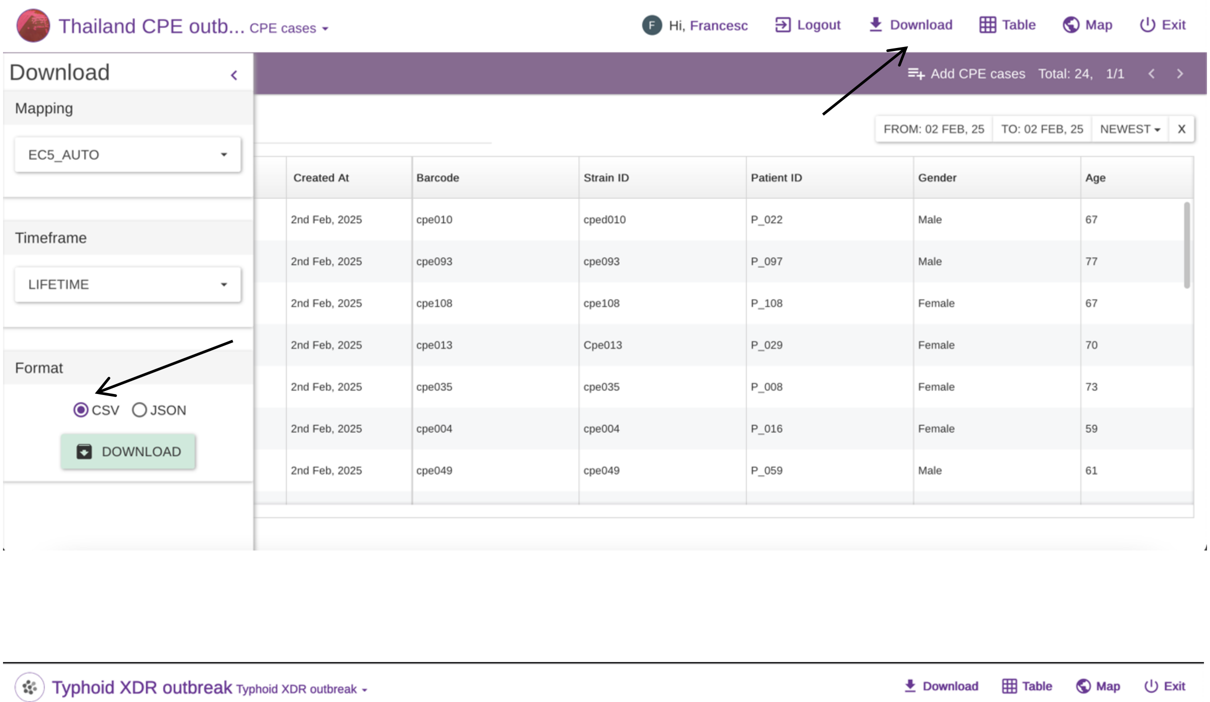

Figure 3 Download EpiCollect entries

Unzip the file 'south-africa-cpe-outbreak-csv.zip' and open it with Excel. We will need to edit this file first to be able to read with MicroReact. Insert a new column on the left-hand side of the Excel spreadsheet. Right-click on the first column and select 'Insert' as show in Figure 4.

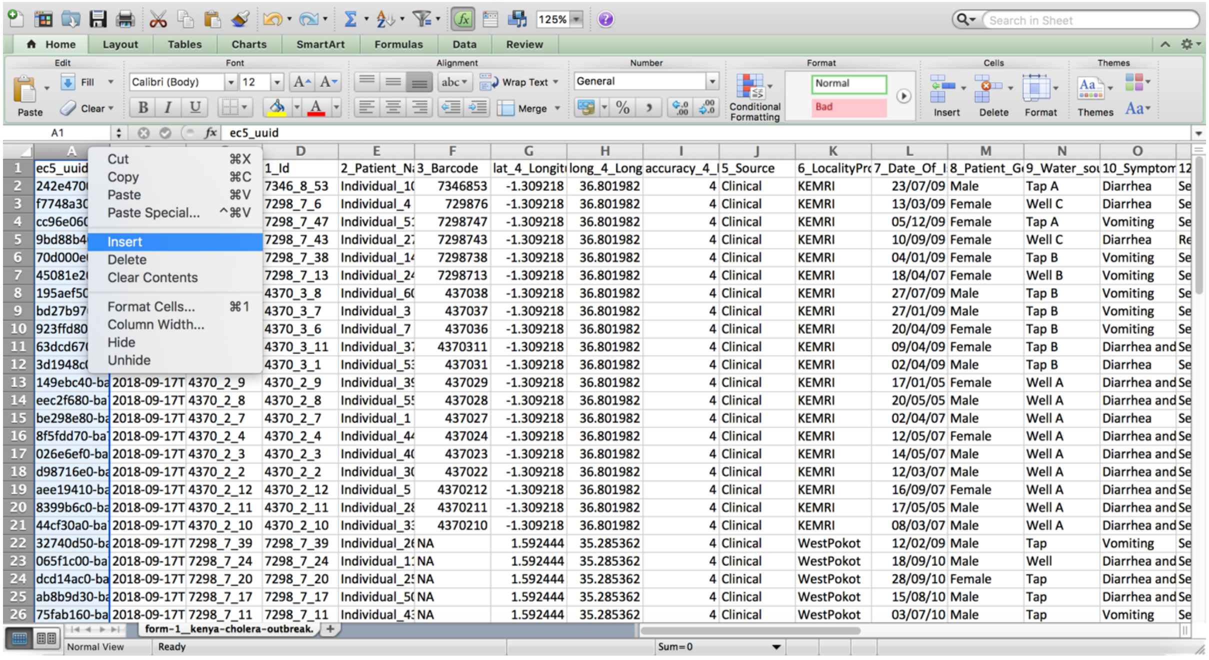

Figure 4 Editing EpiCollect entries using Excel

Cut the column '1_Barcode' and paste it on the newly inserted column as shown in Figure 5.

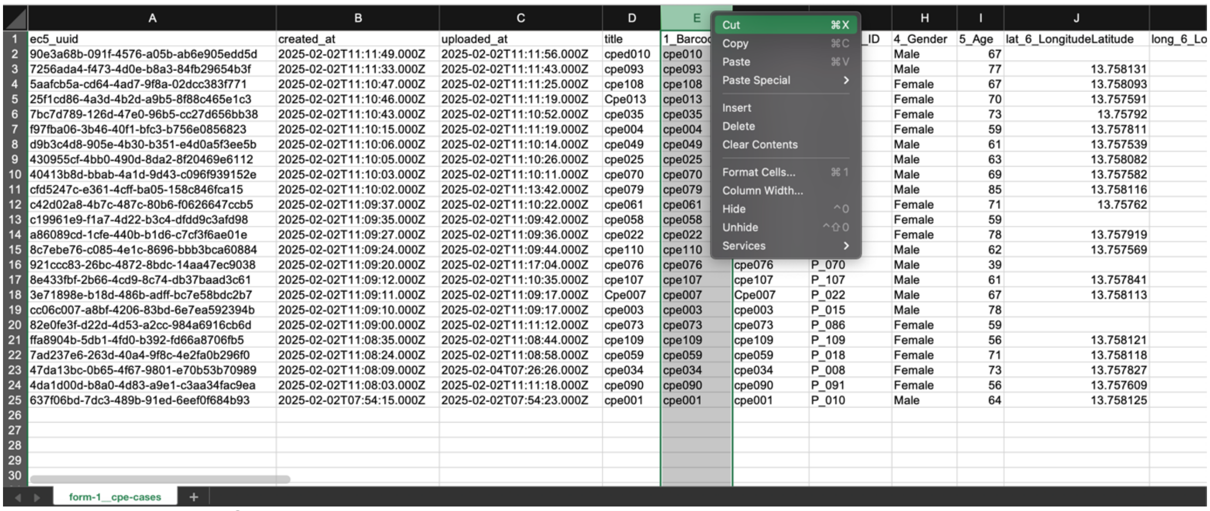

Figure 5 Editing EpiCollect entries using Excel

Change the name of the first column to 'Id' and delete the empty column (Figure 6).

	A	B	C	D	E	F	G	H	I	J	
1	Id	ec5_uuid	created_at	uploaded_at	title	2_Strain_ID	3_Patient_ID	4_Gender	5_Age	lat_6_LongitudeLatitude	long_6_Lo
2	cpe010	90e3a68b-091f-4576-a05b-ab6e905edd5d	2025-02-02T11:11:49.000Z	2025-02-02T11:11:56.000Z	cped010	cped010	P_022	Male	67		
3	cpe093	7256ada4-f473-4d0e-b8a3-84fb29654b3f	2025-02-02T11:11:33.000Z	2025-02-02T11:11:43.000Z	cpe093	cpe093	P_097	Male	77	13.758131	
4	cpe108	5aafcb5a-cd64-4ad7-9f8a-02dcc383f771	2025-02-02T11:10:47.000Z	2025-02-02T11:11:25.000Z	cpe108	cpe108	P_108	Female	67	13.758093	
5	cpe013	25f1cd86-4a3d-4b2d-a9b5-8f88c465e1c3	2025-02-02T11:10:46.000Z	2025-02-02T11:11:19.000Z	Cpe013	Cpe013	P_029	Female	70	13.757591	
6	cpe035	7bc7d789-126d-47e0-96b5-cc27d656bb38	2025-02-02T11:10:43.000Z	2025-02-02T11:10:52.000Z	cpe035	cpe035	P_008	Female	73	13.75792	
7	cpe004	f97fba06-3b46-40f1-bfc3-b756e0856823	2025-02-02T11:10:15.000Z	2025-02-02T11:11:19.000Z	cpe004	cpe004	P_016	Female	59	13.757811	
8	cpe049	d9b3c4d8-905e-4b30-b351-e4d0a5f3ee5b	2025-02-02T11:10:06.000Z	2025-02-02T11:10:14.000Z	cpe049	cpe049	P_059	Male	61	13.757539	
9	cpe025	430955cf-4bb0-490d-8da2-8f20469e6112	2025-02-02T11:10:05.000Z	2025-02-02T11:10:26.000Z	cpe025	cpe025	P_039	Male	63	13.758082	
10	cpe070	40413b8d-bbab-4a1d-9d43-c096f939152e	2025-02-02T11:10:03.000Z	2025-02-02T11:10:11.000Z	cpe070	cpe070	P_084	Male	69	13.757582	
11	cpe079	cfd5247c-e361-4cff-ba05-158c846fca15	2025-02-02T11:10:02.000Z	2025-02-02T11:13:42.000Z	cpe079	cpe079	P_071	Male	85	13.758116	
12	cpe061	c42d02a8-4b7c-487c-80b6-f0626647ccb5	2025-02-02T11:09:37.000Z	2025-02-02T11:10:22.000Z	cpe061	cpe061	P_018	Female	71	13.75762	
13	cpe058	c19961e9-f1a7-4d22-b3c4-dfdd9c3afd98	2025-02-02T11:09:35.000Z	2025-02-02T11:09:42.000Z	cpe058	cpe058	P_019	Female	59		
14	cpe022	a86089cd-1cfe-440b-b1d6-c7cf3f6ae01e	2025-02-02T11:09:27.000Z	2025-02-02T11:09:36.000Z	cpe022	cpe022	P_036	Female	78	13.757919	
15	cpe110	8c7ebe76-c085-4e1c-8696-bbb3bca60884	2025-02-02T11:09:24.000Z	2025-02-02T11:09:44.000Z	cpe110	cpe110	P_110	Male	62	13.757569	
16	cpe076	921ccc83-26bc-4872-8bdc-14aa47ec9038	2025-02-02T11:09:20.000Z	2025-02-02T11:17:04.000Z	cpe076	cpe076	P_070	Male	39		
17	cpe107	8e433fbf-2b66-4cd9-8c74-db37baad3c61	2025-02-02T11:09:12.000Z	2025-02-02T11:10:35.000Z	cpe107	cpe107	P_107	Male	61	13.757841	
18	cpe007	3e71898e-b18d-486b-adff-bc7e58bdc2b7	2025-02-02T11:09:11.000Z	2025-02-02T11:09:17.000Z	Cpe007	Cpe007	P_022	Male	67	13.758113	
19	cpe003	cc06c007-a8bf-4206-83bd-6e7ea592394b	2025-02-02T11:09:10.000Z	2025-02-02T11:09:17.000Z	cpe003	cpe003	P_015	Male	78		
20	cpe073	82e0fe3f-d22d-4d53-a2cc-984a6916cb6d	2025-02-02T11:09:00.000Z	2025-02-02T11:11:12.000Z	cpe073	cpe073	P_086	Female	59		
21	cpe109	ffa8904b-5db1-4fd0-b392-fd66a8706fb5	2025-02-02T11:08:35.000Z	2025-02-02T11:08:44.000Z	cpe109	cpe109	P_109	Female	56	13.758121	
22	cpe059	7ad237e6-263d-40a4-9f8c-4e2fa0b296f0	2025-02-02T11:08:24.000Z	2025-02-02T11:08:58.000Z	cpe059	cpe059	P_018	Female	71	13.758118	
23	cpe034	47da13bc-0b65-4f67-9801-e70b53b70989	2025-02-02T11:08:09.000Z	2025-02-04T07:26:26.000Z	cpe034	cpe034	P_008	Female	73	13.757827	
24	cpe090	4da1d00d-b8a0-4d83-a9e1-c3aa34fac9ea	2025-02-02T11:08:03.000Z	2025-02-02T11:11:18.000Z	cpe090	cpe090	P_091	Female	56	13.757609	
25	cpe001	637f06bd-7dc3-489b-91ed-6eef0f684b93	2025-02-02T07:54:15.000Z	2025-02-02T07:54:23.000Z	cpe001	cpe001	P_010	Male	64	13.758125	
26											
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Figure 6 Editing EpiCollect entries using Excel

Change the name of column ‘lat_6_LongitudeLatitude’ to ‘Latitude’ and ‘long_6_LongitudeLatitude’ to ‘Longitude’ (Figure 7). Save your changes and close.

	D	E	F	G	H	I	J	K	L	M	
1	_at	title	2_Strain_ID	3_Patient_ID	4_Gender	5_Age	Latitude	Longitude	accuracy_6_LongitudeLatitude	UTM_Northing_6_LongitudeLatitude	UTM_Easting_6_
2	02T11:11:56.000Z	cped010	cped010	P_022	Male	67					
3	02T11:11:43.000Z	cpe093	cpe093	P_097	Male	77	13.758131	100.536403	18	1521506	
4	02T11:11:25.000Z	cpe108	cpe108	P_108	Female	67	13.758093	100.536418	9	1521502	
5	02T11:11:19.000Z	Cpe013	Cpe013	P_029	Female	70	13.757591	100.536285	39	1521446	
6	02T11:10:52.000Z	cpe035	cpe035	P_008	Female	73	13.75792	100.536427	37	1521483	
7	02T11:11:19.000Z	cpe004	cpe004	P_016	Female	59	13.757811	100.536324	39	1521471	
8	02T11:10:14.000Z	cpe049	cpe049	P_059	Male	61	13.757539	100.536095	30	1521441	
9	02T11:10:26.000Z	cpe025	cpe025	P_039	Male	63	13.758082	100.536317	15	1521501	
10	02T11:10:11.000Z	cpe070	cpe070	P_084	Male	69	13.757582	100.536306	46	1521446	
11	02T11:13:42.000Z	cpe079	cpe079	P_071	Male	85	13.758116	100.536378	8	1521505	
12	02T11:10:22.000Z	cpe061	cpe061	P_018	Female	71	13.75762	100.536284	46	1521450	
13	02T11:09:42.000Z	cpe058	cpe058	P_019	Female	59					
14	02T11:09:36.000Z	cpe022	cpe022	P_036	Female	78	13.757919	100.536424	34	1521483	
15	02T11:09:44.000Z	cpe110	cpe110	P_110	Male	62	13.757569	100.53622	52	1521444	
16	02T11:17:04.000Z	cpe076	cpe076	P_070	Male	39					
17	02T11:10:35.000Z	cpe107	cpe107	P_107	Male	61	13.757841	100.536259	13	1521474	
18	02T11:09:17.000Z	Cpe007	Cpe007	P_022	Male	67	13.758113	100.53638	6	1521504	
19	02T11:09:17.000Z	cpe003	cpe003	P_015	Male	78					
20	02T11:11:12.000Z	cpe073	cpe073	P_086	Female	59					
21	02T11:08:44.000Z	cpe109	cpe109	P_109	Female	56	13.758121	100.536395	15	1521505	
22	02T11:08:58.000Z	cpe059	cpe059	P_018	Female	71	13.758118	100.536387	14	1521505	
23	04T07:26:26.000Z	cpe034	cpe034	P_008	Female	73	13.757827	100.536359	20	1521473	
24	02T11:11:18.000Z	cpe090	cpe090	P_091	Female	56	13.757609	100.536176	24	1521448	
25	02T07:54:23.000Z	cpe001	cpe001	P_010	Male	64	13.758125	100.536487	11	1521506	
26											
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Figure 7 Editing EpiCollect entries using Excel

Data visualization with Microreact

Start by opening a new window in Firefox and typing https://microreact.org/ in the address bar. Click on "Upload" and browse for the phylogenetic tree (Kpn_ST78.cpe058.rmRCB_iqtree.contree.tree) and metadata files (cpe_cases.epicollect_data.csv) to create a new MicroReact project (Figure 8 and 9).

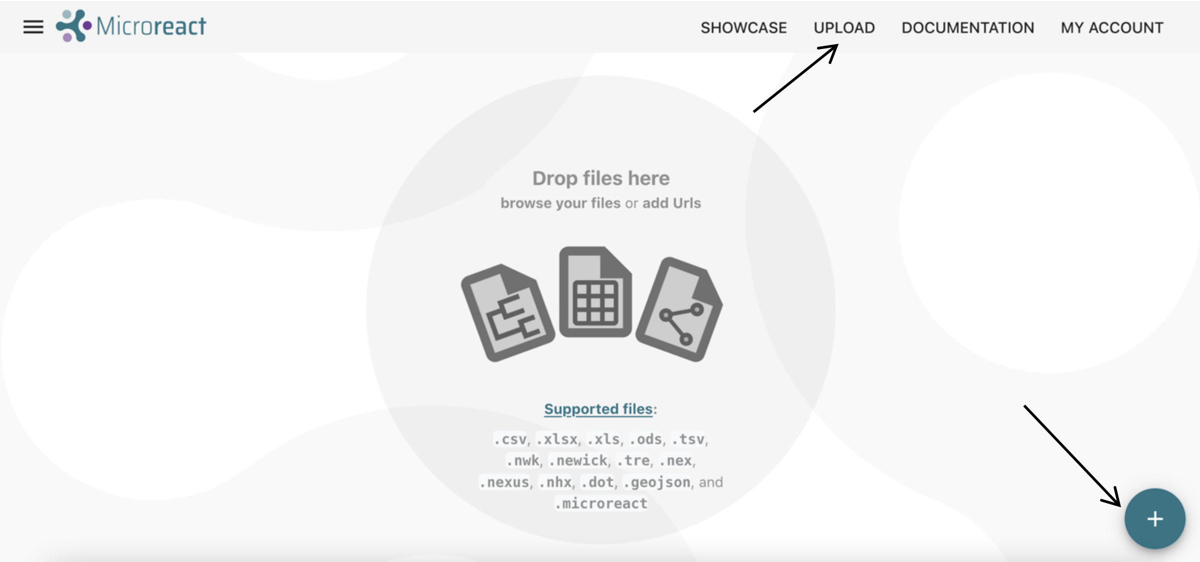

Figure 8 Uploading files on Microreact

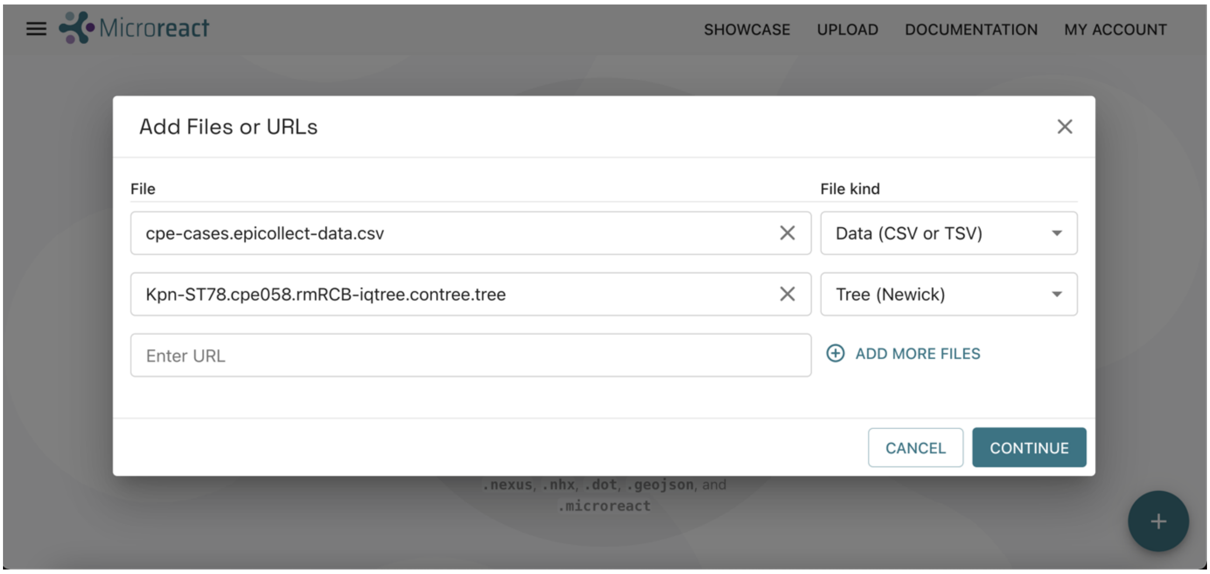

Figure 9 Uploading files on Microreact

Once the tree and metadata files are loaded you will be directed to a new window where files will be automatically detected as Data (CSV or TSV) file (cpe_cases.epicollect_data.csv) and Tree (Newick) file (Kpn_ST78.cpe058.rmRCB_iqtree.contree.tree). In this new window click on 'Continue'. In the next window, make sure the column 'barcode' is selected as the 'ID Column' (Figure 10) and then click on 'Continue'. The 'ID column' is the column in the metadata file that must match the strain labels in the phylogenetic tree (i.e. tip or leave labels).

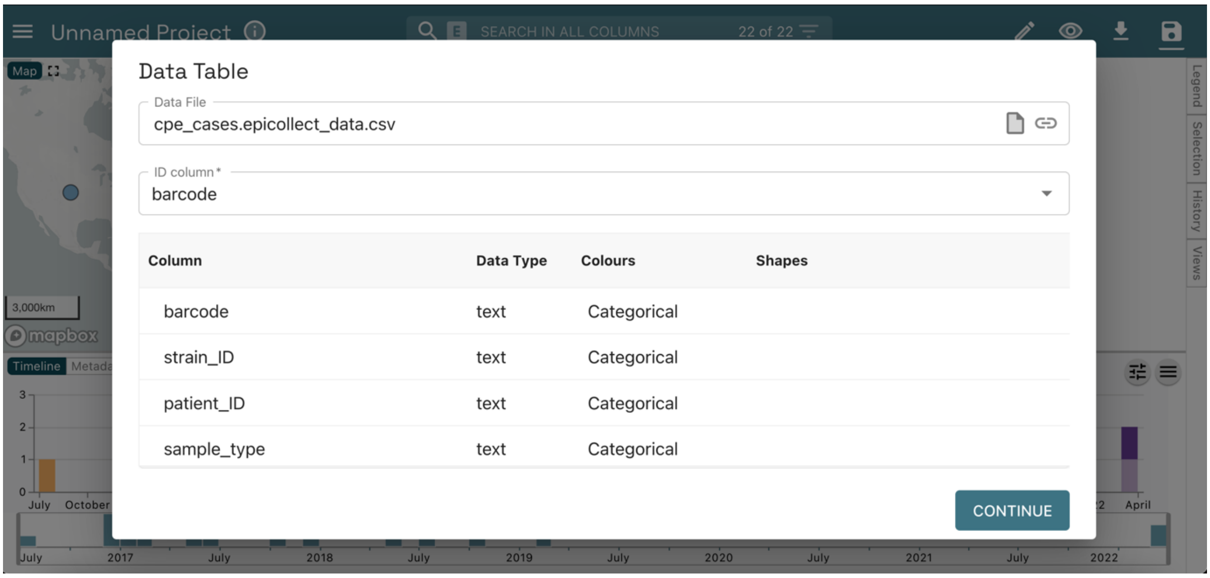

Figure 10 Selecting ID column in Microreact

Once the form is completed your data will be utilized to create a MicroReact project. You should now have a view like the one shown in Figure 11.

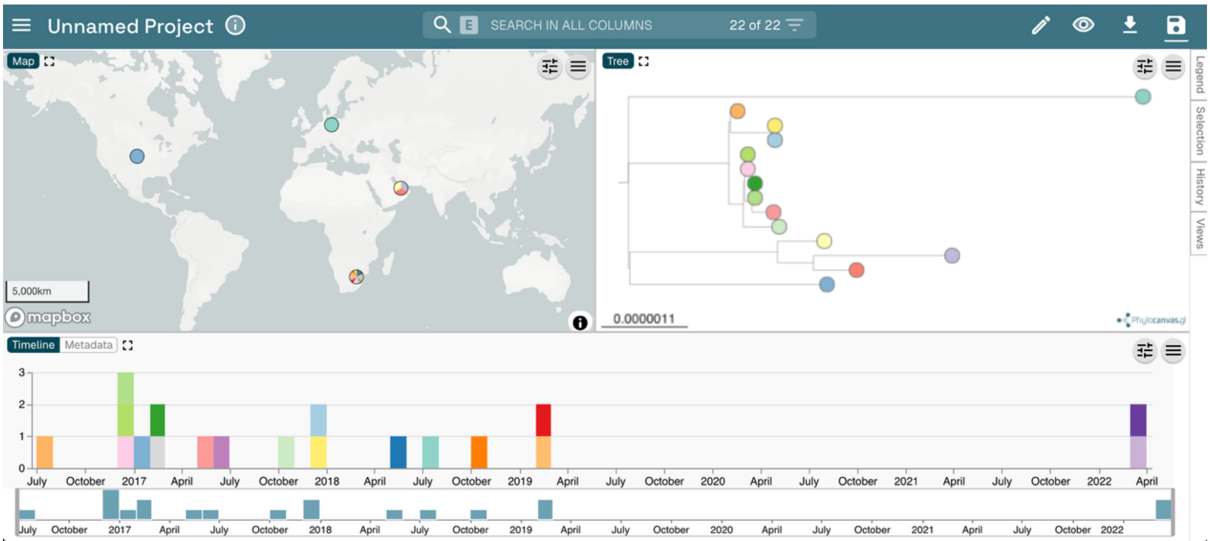

Figure 11 South Africa CPE cases on Microreact

You should see a Map, Tree, and Timeline panels. You can use click-drag-zoom to navigate both the tree and the map.

You can reorient the tree in different layouts to make it easier to view (Figure 12). Click the control panel symbol then click the tree-view control button then select the 'Hierarchical tree' button (the last option at the bottom).

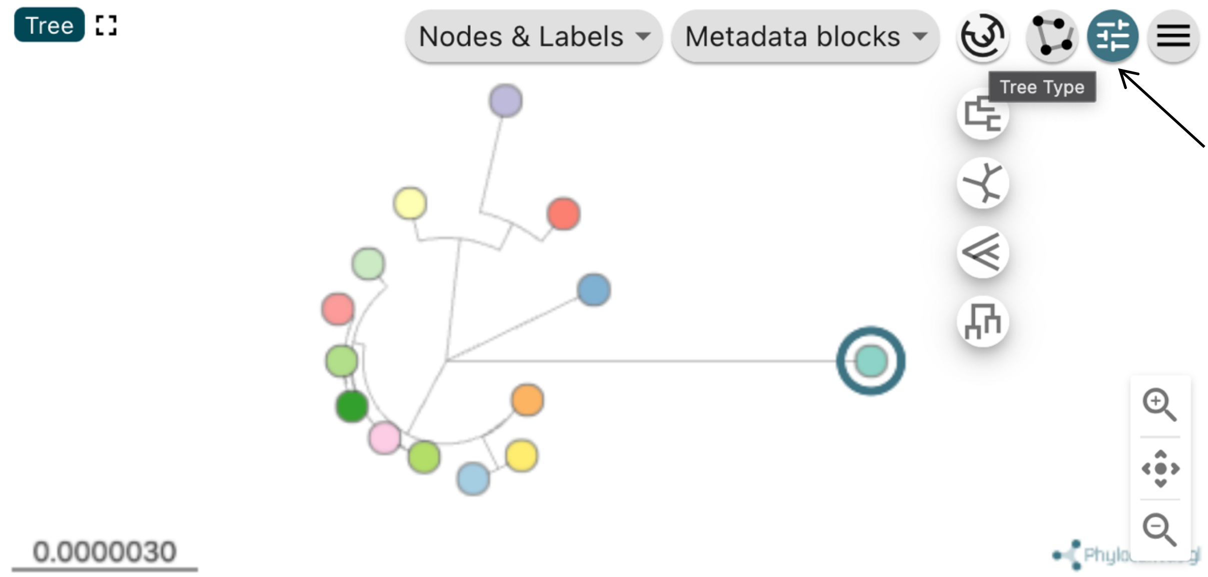

Figure 12 Changing the phylogenetic tree layout on Microreact.

By default, the tree tips should be coloured by strain_ID. You can click on the 'Legend' button to view the colour legend.

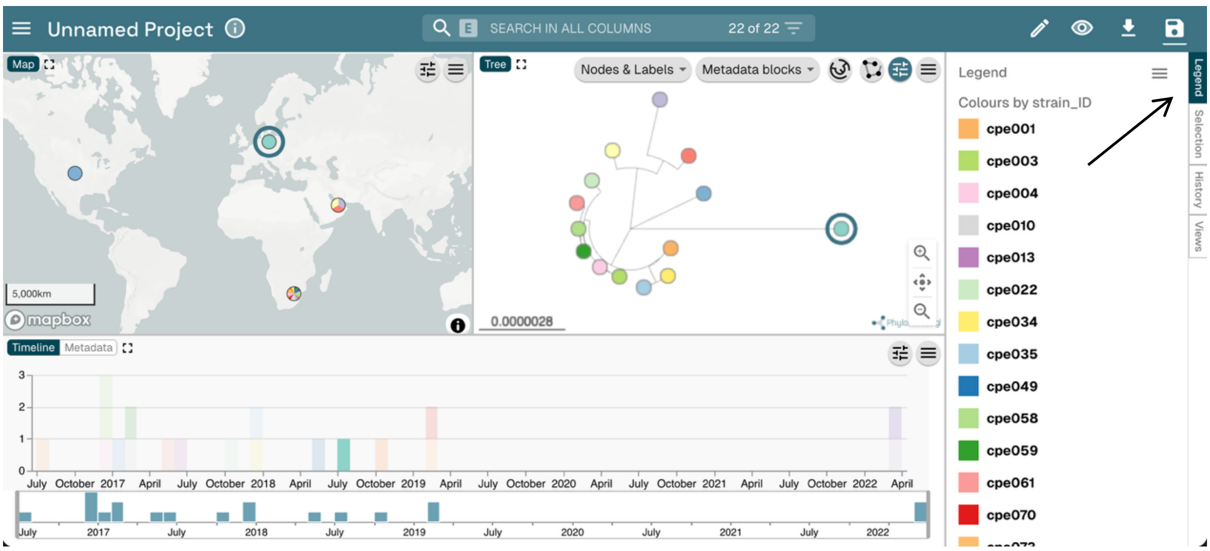

Figure 13 Changing the phylogenetic tree layout on Microreact.

Next, we will also display the strain IDs on the phylogenetic tree. Click on the 'Labels, Colours, and Shapes' icon and select 'strain_ID' under the lists 'Labels Column' and 'Colour Column' (arrows in Figure 14). You can also click the 'Legend' tab (top right) in the Microreact window to see a legend.

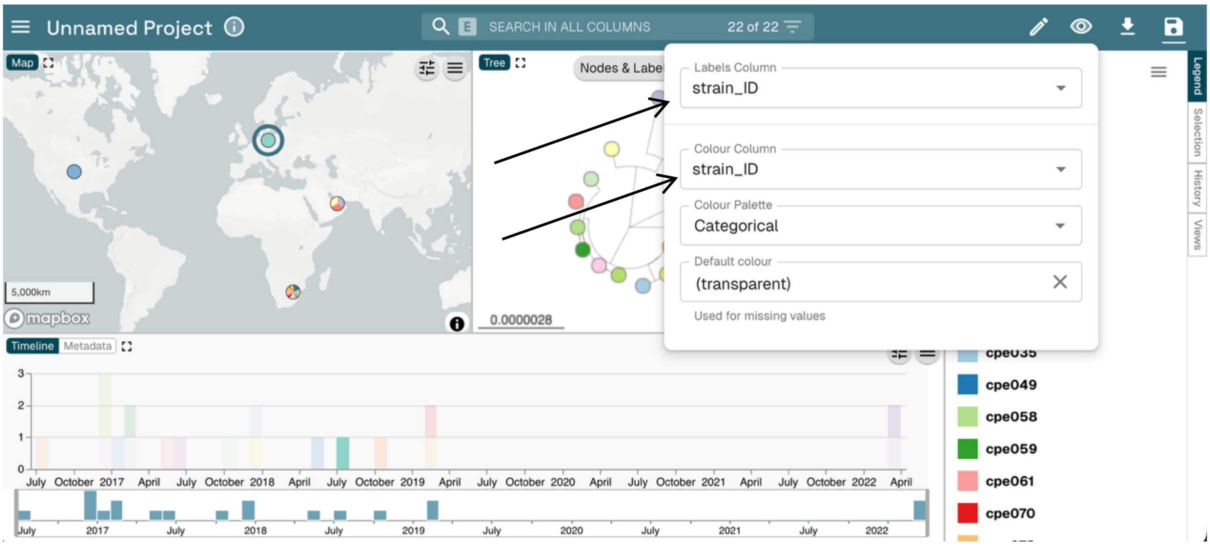

Figure 14 Changing the tip labels and colours.

You should be able to see the clustering of *K. pneumoniae* ST78 isolates with respect to the contextual ST78 isolates (Figure 15).

Question: is there any phylogenetic evidence of a **single-source outbreak** among the *K. pneumoniae* ST78 isolates we are investigating? Or multiple circulating clones? Consider too the SNP distances calculated by *pairsnp* or *MEGA*. Is there any evidence of phylogenetic clustering based on geography?

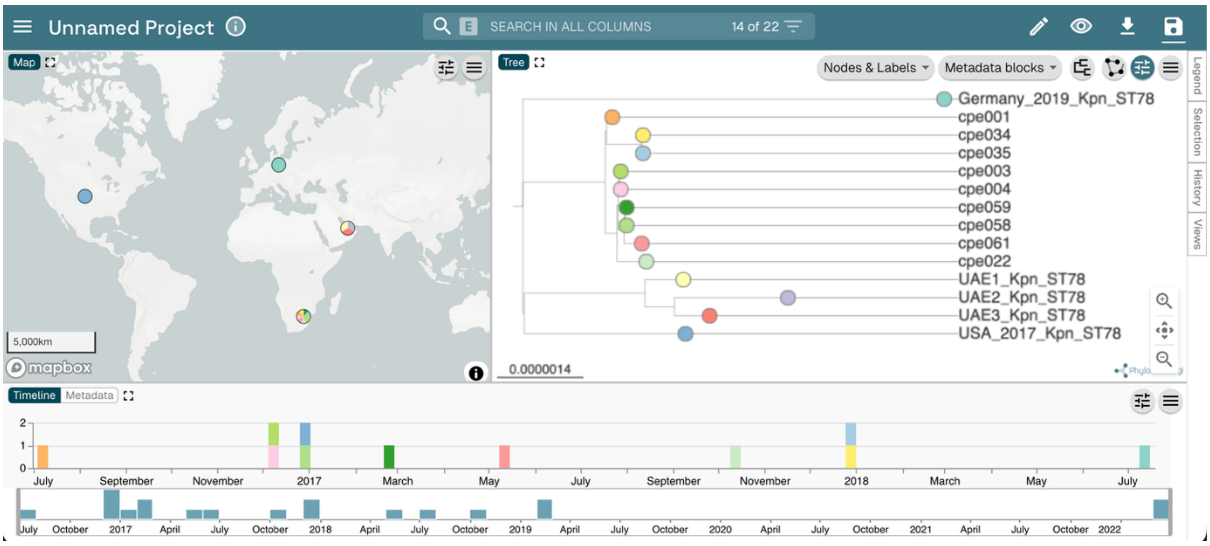

Figure 15 Phylogenetic clustering.

Next, click on the 'Labels, Colours, and Shapes'  icon and select 'patient_ID' under the lists 'Labels Column' and 'Colour Column' (arrows 1 and 2 in Figure 16). You can also click the 'Legend' tab (top right) in the Microreact window to see a legend.

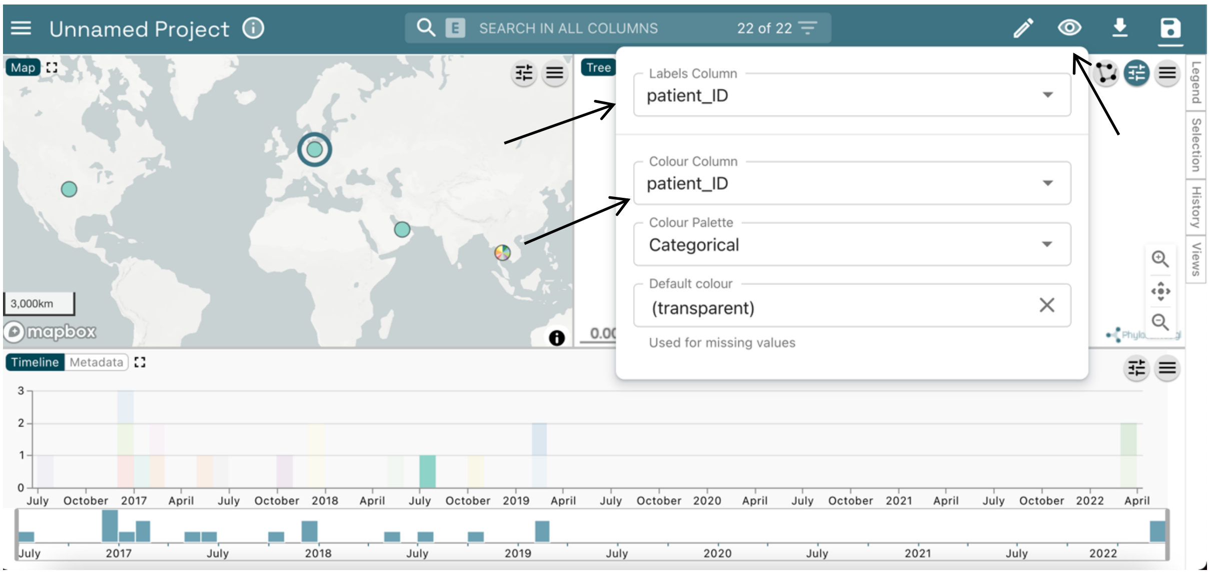

Figure 16 Changing the tip labels and colours.

Click on the 'Show controls' button (Figure 17) to then click the 'Nodes & Labels' menu (above the tree) and toggle the 'Leaf Labels' button to switch on labelling (arrow 2).Unselect 'Align Leaf Labels' (Figure 17).

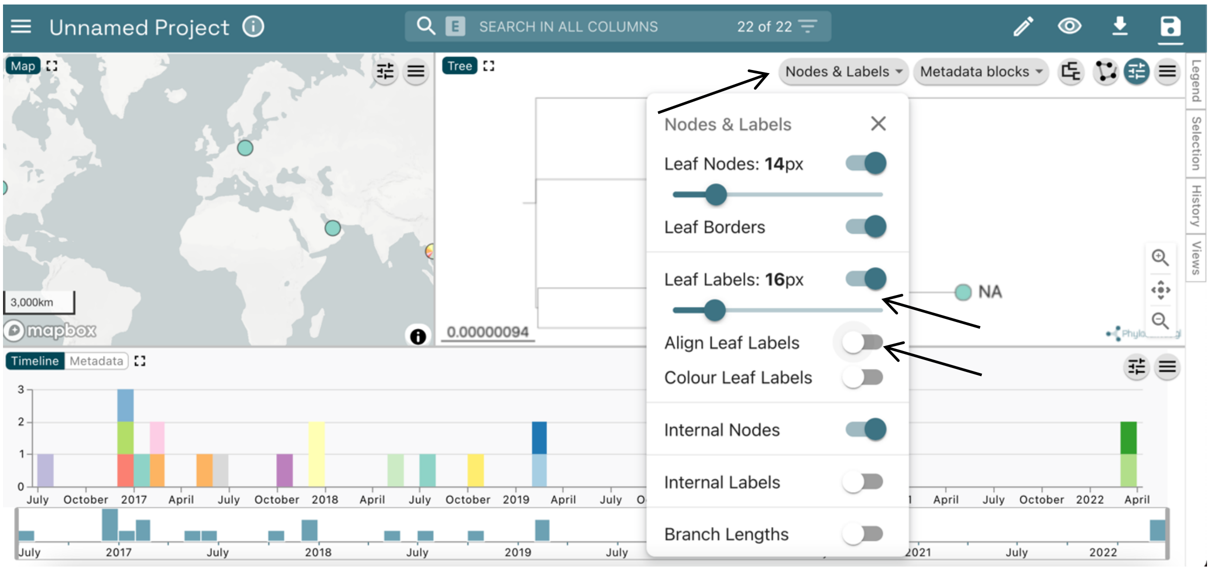

Figure 17 Showing tip labels on the tree.

Question: can you identify any patient with more than one isolate on the phylogenetic tree? (note: ignore the contextual isolates with 'NA' patient_ID, i.e., those with not available patient_ID, which happen to have the same colour)

Next, zoom into the clade of CPE outbreak cases and identify the clustering (sub-clades) of patients. Identify which patients have the most ancestral isolates in each of these sub-clades and consider the sampling time of CPE cases (visible in the Timeline panel, Figure 18).

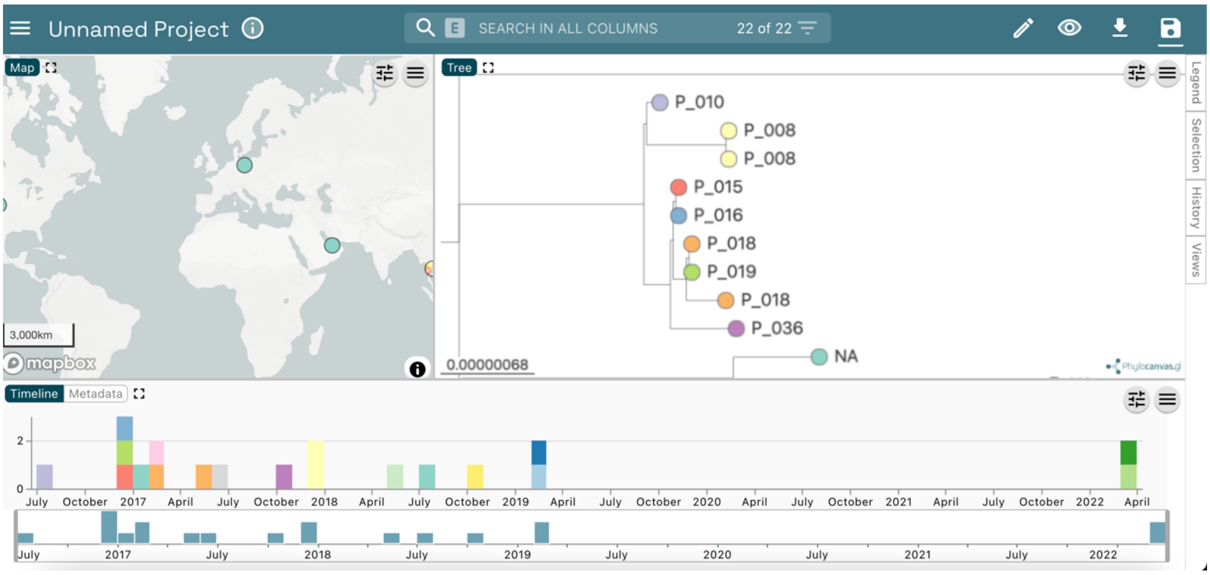

Figure 18 Phylogenetic and temporal clustering of CPE cases.

Question: based on the temporal distribution and phylogenetic clustering of CPE cases, which patients might be among the **early sources of the outbreak**? Which patients could be ruled out as sources of the outbreak? Important note: we cannot infer the origin of an outbreak based solely on phylogenetic and temporal data, we will need epidemiological data too (i.e., hospital contacts, hospital room stays, etc.).

Finally, we will explore the distribution of antibiograms and AMR genetic determinants.

It is possible to view multiple variables against the tree simultaneously. First, click on the 'Show controls' button and under 'Nodes & Labels' menu to toggle both the 'Leaf Labels' and 'Align Lead Labels' buttons to switch on and make sure leaf labels are visible and aligned. Next, click the 'Metadata blocks' menu button and check all the boxes that relate to phenotypic antibiotic resistance (that is, from amikacin to piperacillin_tazobactam). See Figure 19.

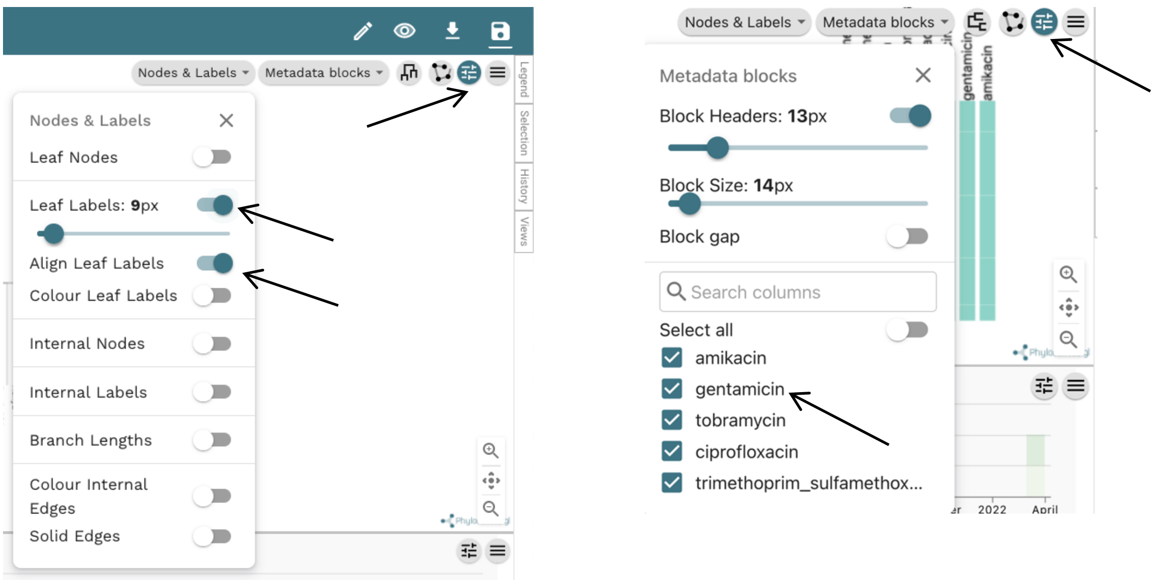

Figure 19 Options to display metadata block.

This will allow us to visualise the entire antibiogram of our local CPE cases and that of contextual strains (Figure 20).

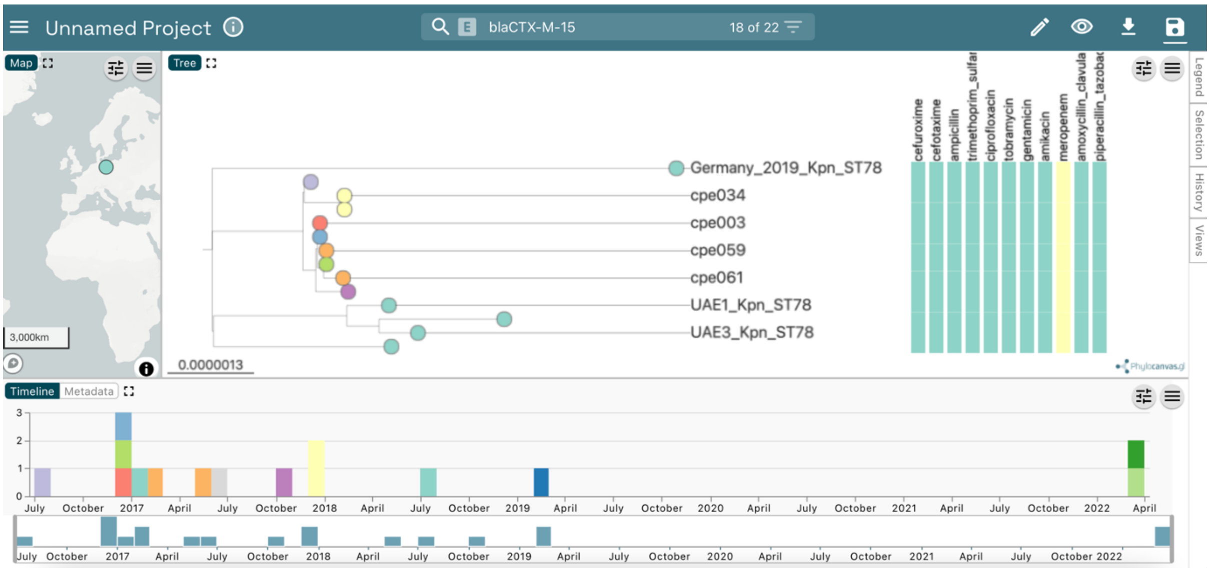

Figure 20 Displaying the antibiogram of CPE cases.

Label the tips of samples with AMR genetic determinants while displaying phenotypic AMR in the metadata blocks. See Figures 21 and 22 as examples.

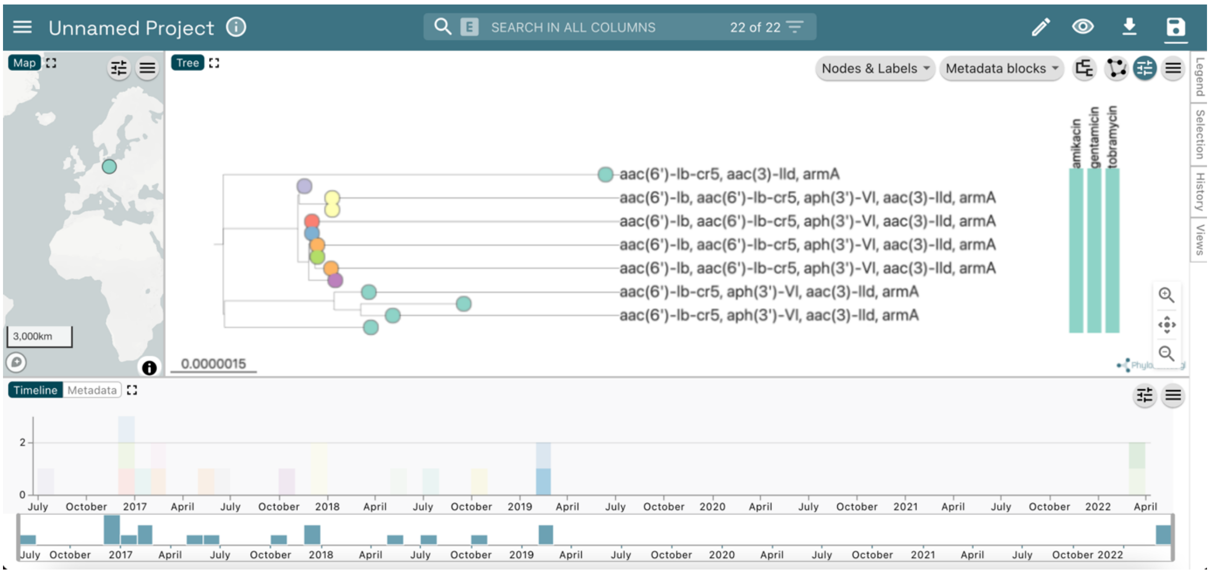

Figure 21 Displaying aminoglycoside resistance genes and resistance phenotypes.

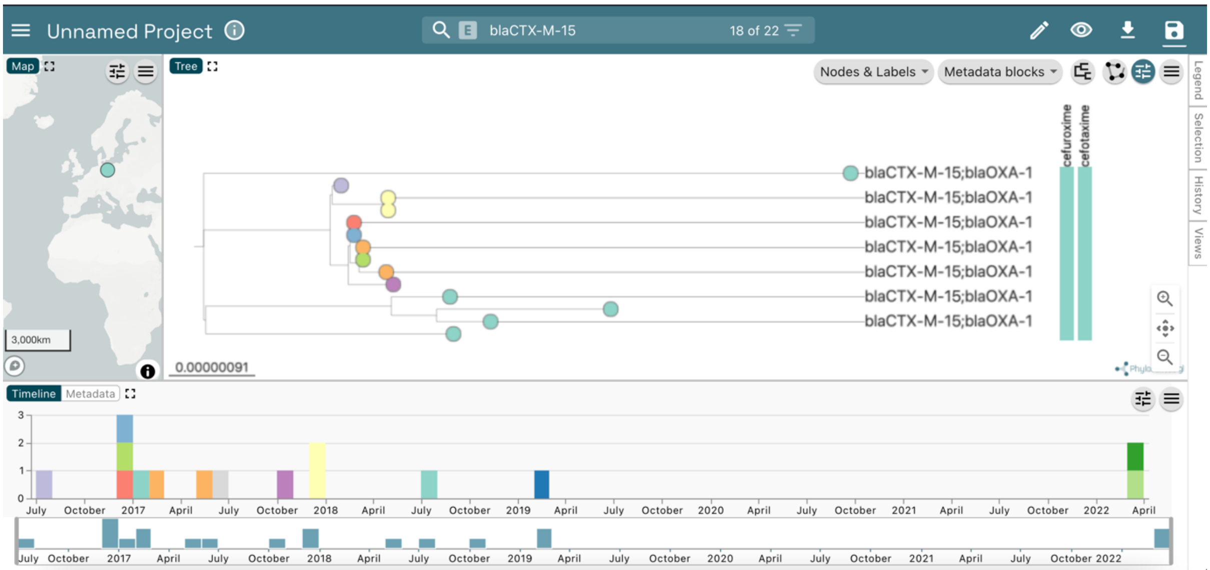

Figure 22 Displaying ESBL resistance genes and resistance phenotypes.

Spend some time exploring the distribution of AMR genes and their relationship with phenotypic resistance.

Question: can you identify any AMR genetic determinants specific to the local CPE cases, and which are shared with the contextual strains?